CHEVALLEY'S THEOREM FOR MORPHISMS OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement carries an optional name argument ('Chevalley's Theorem'); the optional argument is part of the counted statement span one line below the environment start and is preserved verbatim.

Editorial note. The four references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Etale local constructibility tested on affine etale neighbourhoods. Properties of Spaces, Definition spaces-properties-definition-locally-constructible, used at proof lines 5946-5950. Reduces the claim to constructibility of a preimage in an affine V.
- Affine etale covers of quasi-compact algebraic spaces. Properties of Spaces, Lemma spaces-properties-lemma-quasi-compact-affine-cover, used at proof lines 5951-5954. Produces the affine scheme U over V x_Y X.
- The description of points of a fibre product. Properties of Spaces, Lemma spaces-properties-lemma-points-cartesian, used at proof lines 5955-5957. Identifies the preimage of f(E) in V with the image under U -> V of the preimage of E in U.
- Chevalley's theorem for schemes. Morphisms, Lemma morphisms-lemma-chevalley, used at proof lines 5958-5959. Is the scheme case applied to U -> V.

Source reference. Original lemma 054J: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6642–6642. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original definition 0ECU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 17969–17969. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03H4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4553–4553. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 03H6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4555–4555. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Theorem 0.2 (Chevalley's Theorem)**.** *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is quasi-compact and locally of finite presentation. Then the image of every étale locally constructible subset of $|X|$ is an étale locally constructible subset of $|Y|$.*

Proof. Let $E \subset |X|$ be étale locally constructible. Let $V \to Y$ be an étale morphism with V affine. It suffices to show that the inverse image of $f(E)$ in V is constructible, see Properties of Spaces, Definition 0ECU. Since f is quasi-compact $V \times_Y X$ is a quasi-compact algebraic space. Choose an affine scheme U and a surjective étale morphism $U \to V \times_Y X$ (Properties of Spaces, Lemma 03H6). By Properties of Spaces, Lemma 03H4 the inverse image of $f(E)$ in V is the image under $U \to V$ of the inverse image of E in U. Thus the result follows from the case of schemes, see Morphisms, Lemma 054J. □